RECOMMENDER SYSTEMS · BOARDY AI × RECSYS COURSE · 2026

Predicting Introduction Acceptance

40+ experiments on what actually moves the needle when two people have already been matched.

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Full report (GitHub) ↗ Project story ↗

01 · THE TASK

Every pair was already recommended. The label is what humans did next.

200 real introductions — 100 accepted, 100 declined. Every pair passed production's own relevance filter. We're not modeling topic match. We're modeling residual fit: will these two people actually connect?

THE CONSTRAINT

<100 ms latency. Bi-encoders only — candidates pre-embedded, one live query encode + nearest-neighbor lookup.

02 · DATA

200 real pairs. Split user-disjointly. All-200 is the canonical leaderboard.

131 train / 69 held-out, frozen. All main results use all 200 pairs: 100 positive queries ranked against 178 unique candidates. The small held-out set ranks weak models reliably but is uncorrelated with the all-200 ranking among the top models.

EVALUATION RULE

Never make a decision from the small held-out set alone. Re-measure on all 200 before calling anything a win.

02 · METRICS

Two questions. Two metric families. They don't always agree.

Pair AUC — can the model rank accepted above declined? Sliced by hard negatives (high word overlap, the production-relevant population) and easy negatives (low overlap, already filtered by production).

Retrieval — does the accepted candidate appear near the top of 178 candidates? MRR, R@1, R@10.

THE DIAGNOSTIC

Every strong embedding model drops on hard negatives. The LLM judge is the only exception.

Voyage-4-large (0.5726) is the bar. No open-weight model consistently beats it.

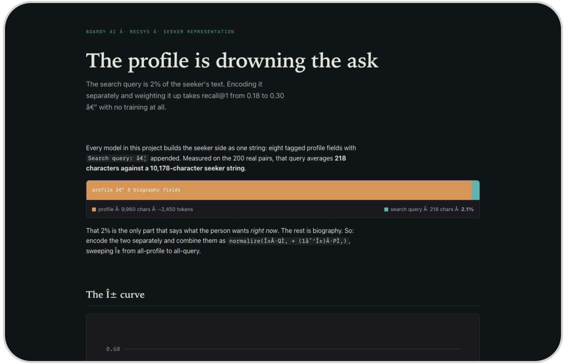
Model	Pair AUC	Hard-neg AUC	MRR	R@1	R@10
Voyage-4-large (production)	**0.5726**	0.5422	0.310	0.13	**0.70**
Voyage-4-nano (frozen)	0.5593	0.5046	0.317	0.18	0.59
TF-IDF	0.5649	0.5164	0.131	0.05	0.26
BGE-en-ICL	0.5389	0.5226	**0.319**	**0.17**	0.62
Qwen3-Embedding-8B	0.5529	0.4680	0.205	0.05	0.55
Frozen BERT	0.4697	0.4108	0.094	0.02	0.18

OBSERVATION

TF-IDF looks strong (0.5649) — but hard-neg AUC is 0.5164, near chance. It is solving yesterday's problem.

The search query is 2% of the text. Separate it and recall nearly doubles.

Approach	Pair AUC	Hard-neg AUC	R@1	R@10
Concatenate (baseline)	0.5593	0.5046	0.18	0.59
Query only (α=1)	0.5530	**0.5914**	**0.30**	**0.91**
Blend α=0.6	**0.5872**	0.5818	0.25	0.89
Profile only (no query)	0.5424	0.4862	0.09	0.50

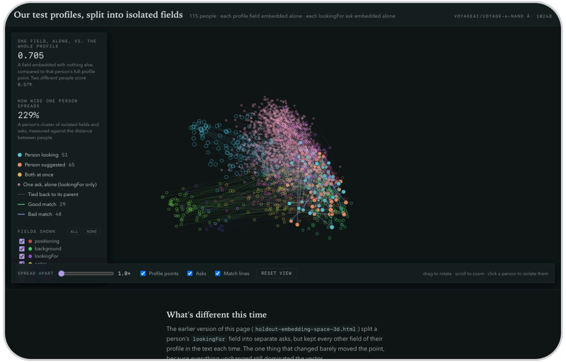


ZERO TRAINING

R@1 0.18→0.30, R@10 0.59→0.91. α=0.6 gives best pair AUC. Pattern replicates on every fine-tuned adapter.

Three fields carry identity. Three fields are scheduling boilerplate.

Field	Own-profile sim	Same field, others
positioning	0.890	0.549
background	0.850	0.580
lookingFor	0.852	0.595
notes	0.700	0.563
introPreferences	0.731	0.680
locationAvailability	0.450	0.676
meetingPreferences	0.391	0.751
personalPreferences	0.366	0.756



DROP THESE

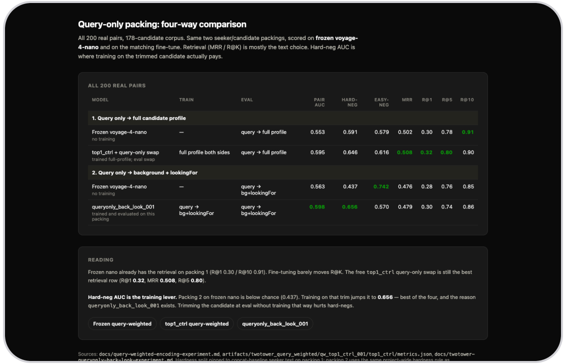
Bottom three fields are more similar to other people's same field than to the person's own profile. Noise, not signal.

04 · BEST TEXT COMBINATION

Seeker = search query only. Candidate = background + lookingFor.

A 105-field-combination sweep confirmed this pairing beat every other combination on the frozen model. It became the basis for the best serving-feasible fine-tune.

It also reduces live encoding work: ~2,500 tokens → ~55 tokens for the seeker.



Query-only packing: four-way comparison

All 200 real pairs, 178-candidate corpus. Same two seeker/candidate packings, scored on **frozen voyage-4-nano** and on the matching fine-tune. Retrieval (MRR / R@K) is mostly the text choice. Hard-neg AUC is where training on the trimmed candidate actually pays.

ALL 200 REAL PAIRS

MODEL	TRAIN	EVAL	PAIR AUC	HARD-NEG	EASY-NEG	MRR	R@1	R@5	R@10
1. Query only → full candidate profile									
Frozen voyage-4-nano no training	—	query → full profile	0.553	0.591	0.579	0.502	0.30	0.78	0.91
top1_ctrl + query-only swap trained full-profile; eval swap	full profile both sides	query → full profile	0.595	0.646	0.616	0.508	0.32	0.80	0.90
2. Query only → background + lookingFor									
Frozen voyage-4-nano no training	—	query → bg+lookingFor	0.563	0.437	0.742	0.476	0.28	0.76	0.85
queryonly_back_look_001 trained and evaluated on this packing	query → bg+lookingFor	query → bg+lookingFor	0.598	0.656	0.570	0.479	0.30	0.74	0.86

READING

Frozen nano already has the retrieval on packing 1 (R@1 0.30 / R@10 0.91). Fine-tuning barely moves R@K. The free `top1_ctrl` query-only swap is still the best retrieval row (R@1 **0.32**, MRR **0.508**, R@5 **0.80**).

Hard-neg AUC is the training lever. Packing 2 on frozen nano is below chance (0.437). Training on that trim jumps it to **0.656** — best of the four, and the reason `queryonly_back_look_001` exists. Trimming the candidate at eval without training that way hurts hard-negs.

Frozen query-weighted · top1_ctrl query-weighted · queryonly_back_look_001

Sources: `docs/query-weighted-encoding-experiment.md`, `artifacts/twotower_query_weighted/qw_top1_ctrl_001/top1_ctrl/metrics.json`, `docs/twotower-`

THE LESSON

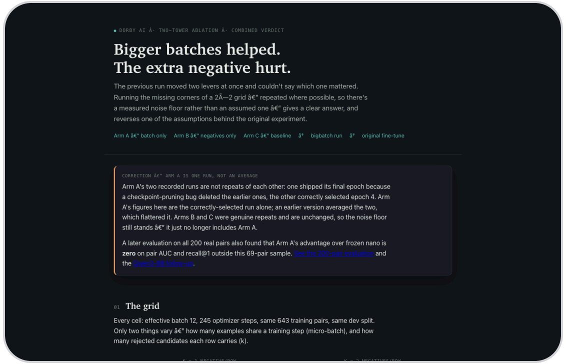
What you feed the encoder at inference consistently outweighed training choices throughout this project.

05 · RESULT 2 — FINE-TUNING

LoRA adapter on a frozen encoder. Batch size is the key training lever.

<1% of parameters trained. Adapter merges into base weights at serving time — no extra overhead. Two base models: **Voyage-4-nano** (serving-feasible) and **Qwen3-8B** (higher accuracy).

Micro-batch 6 beats micro-batch 2 by 5 R@1 points. The gain comes from more candidates per forward pass — gradient accumulation does not replicate it.



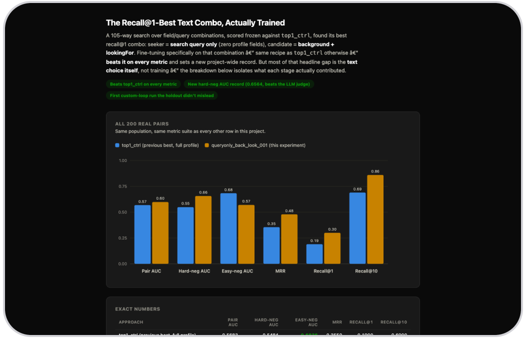
TRAINING INSIGHT

More candidates per forward pass beats gradient accumulation. Every structural addition failed.

05 · FINE-TUNE RESULTS

Qwen3-8B sets the record. Nano on query→bg+lookingFor is the serving-feasible winner.

Model	Pair AUC	Hard-neg AUC	MRR	R@1	R@10
Qwen fine-tune (voyage-gemini) caveat	**0.6446**	**0.6862**	0.415	0.24	—
Nano: query→bg+lookingFor	0.5983	0.6564	0.479	0.30	0.86
Voyage-4-nano (frozen, query-only)	0.5530	0.5914	**0.502**	**0.30**	**0.91**
Voyage-4-large (production)	0.5726	0.5422	0.310	0.13	0.70



KEY FINDING

Nano query→bg+lookingFor: first model where hard-neg AUC (0.6564) exceeds easy-neg AUC — previously only the LLM judge.

The first batch was actively harmful. 99.2% of the label was in the text.

A classifier on the candidate's text alone — no seeker, no query — predicted accept/decline at 99.2% accuracy. The fine-tune trained on this batch scored 0.4845 hard-neg AUC, below chance. The real-only baseline (111 pairs) beat it on every metric with a fifth of the data.

THE FIX

Quarantine the batch. Invent the people first, decide the match later — and verify with three probes before training.

Three cheap checks gate every batch before training.

1. **Candidate-only AUC** — word-frequency on candidate text predicts label? Real floor ~0.49. Above 0.55 = leakage.
2. **Lexical circularity** — query↔candidate word-overlap predicts label? If yes, model learns overlap, not fit.
3. **Seeker-identity AUC** — knowing only which seeker predicts label? If yes, per-seeker base rates dominate.

Batch	Candidate-only AUC	Lexical circ.	Seeker-identity
First batch quarantined	**0.992**	—	—
rrf_002	0.634	0.701	0.687
rrf_003 (main training batch)	—	—	~0.687
voyage-gemini batch leakier	0.758	0.481	0.780

CLEAN PIPELINE

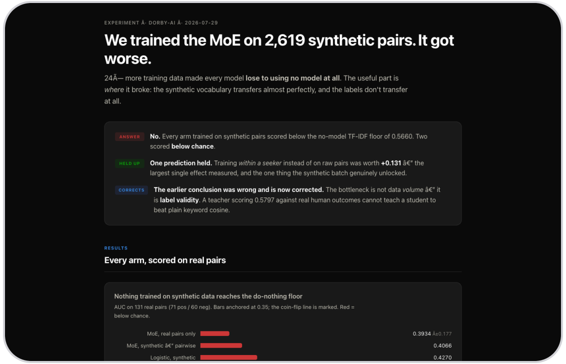
Dense (Qwen3-8B) + BM25 → weighted RRF → Gemini judge. Retrieval and labeling are independent.

06 · WHAT TRANSFERS

Vocabulary transfers. Labels do not.

A model trained to predict generated labels scores **0.427 AUC** on real outcomes — below chance.

The LLM judge scoring ~0.59 on hard pairs is the ceiling for anything trained on its labels. The best fine-tune (0.6446) is approaching that ceiling.

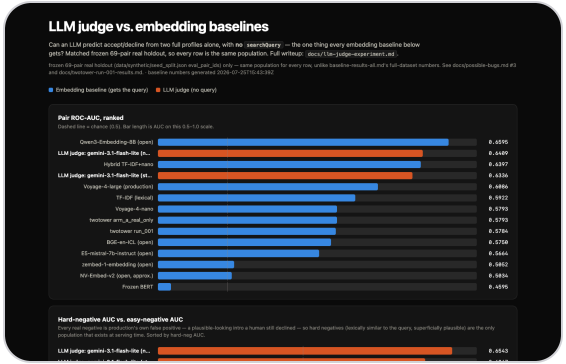


CEILING

More real accept/decline outcomes would unblock progress more directly than new model architectures.

Best overall: 0.6451 AUC. Only approach that does better on hard negatives.

Prompt variant	Pair AUC	Hard-neg	Easy-neg
Focused (trimmed fields + query)	**0.6451**	**0.6590**	0.6570
Naive (complete profiles)	0.6177	—	—
Forced step-by-step CoT	0.6100	0.6225	0.6394
Calibrated (told base rate)	0.5901	0.5879	0.5310



Adding information hurt. Forced reasoning hurt. All 10 automated prompt rewrites scored below the original.

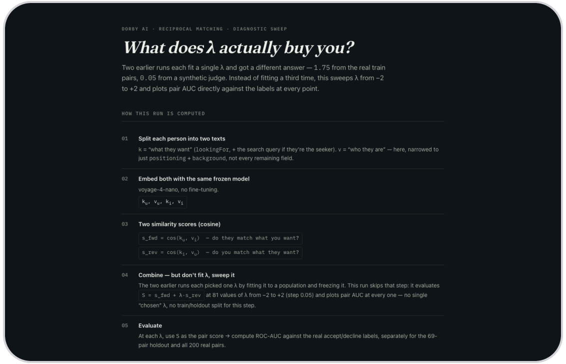
NOT DEPLOYABLE

Per-candidate LLM calls take seconds. Value: accuracy ceiling and training-data labeler.

Score fit in both directions. Positive λ always helps.

```
forward + λ × sim(candidate lookingFor, seeker background)
```

Encoder	Forward AUC	Combined (best λ)	Held-out
Frozen Voyage-4-nano	0.5638	0.5964 (λ=1.75)	0.6241
top1_ctrl fine-tune	0.5890	0.5961 (λ=0.50)	0.6853
voyage_gemini_ctrl caveat	0.6081	**0.6587** (λ=1.90)	**0.7345**



CONSISTENT DIRECTION

Negative λ always hurts (drops to ~0.49 at λ=−2). The reciprocal term carries real directional signal.

Score each lookingFor paragraph separately. +0.071 AUC over logistic regression.

Each pair becomes multiple rows (5.4 paragraphs per seeker on average). A mixture-of-experts model (4 experts, TF-IDF embeddings) scores each row separately and pools. 5×5-fold CV on 131 real train pairs.

Approach	Mean AUC	vs logistic regression
Plain logistic regression	0.5758	—
Per-ask, attention pooling	**0.6467**	+0.071, wins 5/5 seeds
Per-ask, simple average	0.6404	+0.065, wins 5/5 seeds
Single network (no MoE)	0.5446	−0.031

MOST REPLICATED GAIN

Pooling method doesn't matter (attention ≈ average). The mixture of experts matters (+0.10 vs single network). Held-out not yet checked.

Eight additions. All lost to choosing better text.

Addition	Result	What won instead
Two separate encoders	Hard-neg AUC 0.483, below chance	One shared encoder
Learned field selector	Worst retrieval (R@1 0.23)	Fixed blend α=0.6
KL drift penalty	Slightly worse in two separate tests	Plain training
Sharper training objective	R@1 0.18→0.14, below frozen	Default settings
Bilinear score correction	CV 0.74, honest 0.62	Cosine + LSA compression
Forced step-by-step LLM reasoning	AUC 0.6409→0.6336, yes-rate 55%→75%	Naive direct prompt
Automatic prompt rewrites (10 runs)	All 10 below the original	Original prompt

PATTERN

At 200 pairs, complexity is a liability. What you encode beats model architecture every time.

11 · SERVING

Bi-encoders only. No latency benchmark exists yet.

The <100 ms constraint rules out LLM judges, cross-encoders, and remote API calls per candidate. Offline candidate embeddings + one query encode + ANN lookup is the only viable shape. A merged LoRA adapter adds no serving overhead.

OPEN

Whether the best serving-feasible config actually completes in under 100 ms on Boardy's hardware has not been measured. That measurement comes first.

Five threats to every number in this project.

Small held-out set unreliable for strong models

Top-6 ranking on held-out vs all-200: Spearman –0.029. Qwen3-8B reversed from 0.6595 to 0.5529. Use all-200.

131 pairs manufactures results under selection

Bilinear MF: CV 0.74, honest 0.62. A 0.12 gap from 64 configurations. Always multi-seed CV.

Retrieval metrics are partly self-graded

Production selected candidates with the same searchQuery used for evaluation. Pair AUC does not have this circularity.

Generated labels capped by the judge · B-data incomparable

Judge ~0.59 on hard pairs is the ceiling. B-data (46k rows) uses different labels, 21k candidates, ~86%

accepted — incomparable to 200-pair dataset.

What you encode matters more than which model you use.

Query-weighted retrieval: R@1 0.30, R@10 0.91 — frozen model, no training

The biggest single improvement in the project. Replicates on every fine-tuned adapter.

Fine-tune on clean data: 0.6446 AUC, 0.6862 hard-neg AUC

Three leakage probes separate helpful batches from harmful ones. The first batch was actively harmful.

Recommended design: query retrieval → reciprocal re-ranker

Retrieval and accept/decline are different problems. The bottleneck is label quality, not model architecture.

Open: latency · cleaner batch · sectioned MoE holdout check

Nothing has been timed under 100 ms. Arm C (clean full-scale batch) is not yet scheduled.

Full report + appendices ↗ Project story ↗ Experiment index ↗